

# MLE

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2:18 PM



Maximum  
Likelihood...

1-7

⇒ Data follows some distribution  $P(X, Y)$ ,  
↳ If we have the distribution then we can fit optimal Bayes classifier (Most optimal classifier in that data set).  
↳  $P(Y|X)$  is Bayesian optimal classifier.

⇒ Discriminative approach

→  $P(Y|X) \cdot P(X) \Rightarrow$  Given the data what will the label look like.

→  $P(X|Y) \cdot P(Y) \Rightarrow$  Given the label, what will the data look like.

Q) Assume you found an old coin, and you want to know the probability distribution

$$P(X) = H, T, T, H, H, H, T, T, T, T.$$

$$P(H) \approx \frac{n_H}{n_H + n_T}.$$

↳ We want to derive this.

- MLE :- MLE is about finding the parameter value that makes your observed data most likely under the chosen distribution

$$P(D; \theta) \quad , \quad \theta = \operatorname{argmax} P(D; \theta).$$

i.e. The most likely distribution (not type of distribution), s.t. I observe what I observe.

→ we want to find the parameter that maximizes the probability of the observed data.  
 ↳ i.e. Most likely parameter that I observe what I have observed

⇒ ∴ We have two different values for random variable  $X$ . then we have a binomial distribution.

$$\therefore P(D; \theta) = \binom{n_H + n_T}{n_H} \cdot \theta^{n_H} \cdot (1-\theta)^{n_T}$$

↳ Binomial distribution.

$$\therefore \theta = \arg \max_{\theta} P(D; \theta)$$

$$\therefore \theta = \arg \max_{\theta} -\log P(D; \theta).$$

$$\theta = \arg \max_{\theta} \log \binom{n_H + n_T}{n_H} + n_H \log \theta + n_T \log (1-\theta)$$

$$\therefore \frac{dP}{d\theta} = 0 + \frac{n_H}{\theta} - \frac{n_T}{(1-\theta)}$$

$$\therefore (1-\theta)n_H - \theta(n_T) = 0$$

$$\theta = \frac{n_H}{n_H + n_T}$$

⇒ This does not work for small data.

↳ Like I tossed an only one coin toss

↳ to counter this I can assume a prior ex:- 50-50 chance of heads & tails.

$$\theta = \frac{n_H + m_H}{n_H + n_T + m_H + m_T} \rightarrow \text{I have seen } m_H \text{ before}$$

(assumptions).

$$n_H + n_T + m_H + m_T$$

↳ This is called smoothing that frequentists have altered for a long time.

⇒ The more you smooth, the more you're leaning towards your prior belief.

↳ In most application you do +1.

⇒ MLE is frequentist, Maximum A Prior Estimation is Bayesian statistics,   
 ↳ Where we use smoothing.

$P(D; \theta) \Rightarrow$  Frequentist.

$P(D|\theta) \Rightarrow$  Bayesian.

↳ Probability of D given  $\theta$ ,  $\theta$  is a random variable that D can condition.

⇒ Frequentist say  $\theta$  is not probabilistic event, no sample space that  $\theta$  is associated with any outcome.

|     |                                      |                                                             |
|-----|--------------------------------------|-------------------------------------------------------------|
| MLE | $P(D \theta) \leftarrow$ Likelihood. | } $P(\theta D) = \frac{P(D \theta) \cdot P(\theta)}{P(D)}.$ |
|     | $P(\theta) \leftarrow$ Prior.        |                                                             |
| MAP | $P(\theta D) \leftarrow$ Posterior.  |                                                             |

⇒ Bayesians consider that the parameter to a distribution have a distribution of their own distribution ex:- Beta distribution.

$$P(\theta|D) \propto P(D|\theta) P(\theta).$$

$$\binom{n_H+n_T}{n_H} \cdot \theta^{n_H} (1-\theta)^{n_T} \cdot \theta^{\alpha-1} (1-\theta)^{\beta-1}$$

$$\propto \binom{n_H+n_T}{n_H} \theta^{(n_H+\alpha-1)} (1-\theta)^{(n_T+\beta-1)}.$$

If you do MLE on this you get the smoothed result that we got before

$$\theta = \frac{n_H + (\alpha - 1)}{n_H + n_T + (\alpha + \beta - 2)}.$$

↳  $(\alpha - 1 + \beta - 1).$

- Bayesian Approach allows  $\theta$  to be a random variable, & allows for enormous flexibility.

$P(\theta|D) \Rightarrow$  Most likely  $\theta$  given  $D$ .

$$\Rightarrow P(\text{heads}|\text{Data}) = \int_{\theta} P(\text{heads}|\theta, D) d\theta = \int_{\theta} P(\text{heads}|\theta, D) P(\theta|D) d\theta$$

(for every possible theta)

What is the probability that I get heads &  $\theta$  given the data

$\Rightarrow$  allows to integrate out your model.

The Prediction  $P(\text{heads}|D)$  no longer depend on any model

is equal to if I had any of the model in universe multiplied by the probability of seeing that model given that the data we had, and we integrate over all such model to find prediction given the data.



## Gaussian Processes...

⇒ Gaussian distributions is the block hole of distributions.

- Adding a GD with other GD, yields a GD.
- Multiplying  $\dots$
- Marginalizing  $\dots$
- Normalizing  $\dots$
- Conditioning  $\dots$

$$\text{MLE} = P(D|w) = \prod_{i=1}^n P(y_i | x_i; w) / D = \alpha(x_1, y_1) \dots (x_n, y_n)$$

$$\text{MAP: } P(w|D) \propto \frac{P(D|w) \cdot P(w)}{Z} \xrightarrow{\text{Prior}}$$

$$P(y_i | x_i; w) = \mathcal{N}(w^T x_i, \sigma^2 I) \Rightarrow f(x) w^T x + \epsilon$$

$\uparrow$   
 Gaussian  
 with mean  
 $w^T x$  &  $\epsilon$ .

⇒ Instead of model the probability of  $w$ , & then make prediction, why don't we, from start model the prediction of a test point.

Let  $x$  be a test point, &  $D$  be the train Data.

$P(y|x, D)$ .  $\{$  Makes no assumption on model, given that  $x$  is my test point, what's the distribution of my label.

$$P(y|x, D) = \int_{w} P(y|x, w) \cdot P(w|D) dw.$$

